Effect of Minor Ce Additions on Corrosion Behavior of Experimental Pipeline Steel

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ABSTRACT

Minor additions of rare earth elements (REEs) have been shown to improve various steel properties. In particular, Cerium (Ce), as a relatively abundant and inexpensive element, has received attention as a potentially beneficial minor alloying addition for a variety of steel applications, including pipeline materials. The objective of this work is to investigate the effect of Ce on corrosion performance of experimental pipeline steel. Electrochemical corrosion experiments were carried out on steel without and with 0.0057, 0.0164, and 0.0263 wt.% of Ce in 3.5 wt.% NaCl saturated with CO_2 at 20 °C. For comparison purposes, X100 carbon steel specimens machined from an experimental pipe with a yield strength of 731 MPa were tested under the same conditions. The electrochemical techniques employed included: potentiodynamic polarization (PDP), linear polarization resistance (LPR), and electrochemical impedance spectroscopy (EIS). In addition, post-corrosion surface characterization was performed using scanning electron microscopy (SEM) equipped with energy dispersive X-ray spectroscopy (EDS). The corrosion of base metal without Ce was significantly lower (0.15 mm/y) compared to carbon steel X100 (0.9 mm/yr). The electrochemical results showed that adding Ce accelerates corrosion rates compared to base metal without Ce. No pitting corrosion of tested steel was detected.

Keywords: CO_2 corrosion, steel pipelines, rare earth elements, Cerium

INTRODUCTION

High-pressure steel pipeline is a common, cost-effective method for transporting hydrogen and captured carbon dioxide (CO_2) from its point of capture to storage sites [1]. In the past decade, high-strength steel pipeline technology has been developing in response to rapidly increasing energy demand. In general, carbon steels (X60, X65, and X80) are the main commercial materials used in constructing CO_2 transportation pipelines [2]. Higher steel grades are being considered for future use in higher pressure transmission pipelines. However, high-strength steels can experience sudden failure due to hydrogen accumulation in the lattice or in defects, such as dislocations, grain boundaries, and carbides. In addition, steel pipelines are susceptible to corrosion during the transportation of CO_2, due to the presence of moisture and corrosive impurities such as H_2S and O_2.The unique metallurgical microstructure at pipeline welds, which is combined with a synergism of hydrogen and stress, makes the area prone to corrosion, such as hydrogen-induced degradation, resulting in pipeline failure. Therefore, it is important to manufacture a more robust pipeline material for transporting hydrogen and captured CO_2. In addition, in the pipeline transportation system, the steel pipeline microstructure plays an important role in the corrosion process. Therefore, another method to mitigate CO_2 corrosion is by using corrosion-resistant steel pipelines.

To preserve mechanical properties and modify or remove the formation of MnS inclusions in steel, rare earth elements (REEs) can be added to the steel during the manufacturing process [3, 4]. Several studies proved that REEs in the steel can purify the molten steel [3], modify inclusions [5, 6, 7], and improve the structure and properties of steels [6, 8]. It was found that the type and size distribution of inclusions in the steel changes with the difference in added Ce content [5, 6, 7]. Wan et al. [5] observed the formation of $CeAlO_3$-Ce_2O_2S inclusions when the added Ce content in the steel was below 60 ppm. However, the inclusions were mainly Ce_xS_y-Ce_2O_2S when the added Ce content in the steel was 95 ppm. In the past 10 years, Ce has been one of the most investigated rare-earth salts and has been successfully used to control aqueous corrosion of steel [9]. In addition, REEs have been also found to have a positive effect on the corrosion resistance of steel exposed to marine environments [4]. Montemor et al. [9] evaluated the composition and corrosion behavior of galvanized steel treated with Ce in 0.3 wt.% NaCl. The author observed formation of Ce oxide and/or hydroxide on galvanized steel which constrained the cathodic reduction of oxygen and the anodic dissolution of zinc, leading to a decrease of the corrosion rate [9]. Many studies have been conducted to investigate the effects of REE-modified inclusions on the initiation of pitting corrosion in steels [10, 11, 7]. Lijie et al. [10] showed that REE content of 0.0065–0.016 wt% in weathering steel modified inclusion which significantly improved the resistance to pitting corrosion. Zhang et al. [12] observed dissolution of the steel matrix after adding Ce element to stainless steel which caused pitting corrosion to occur.

Very limited studies on the effect of Ce addition on corrosion behavior of steel in CO_2 aqueous environments have been performed. Therefore, in this work, efforts have been made to investigate the effect of minor Ce additions on corrosion behavior of experimental pipeline steel immersed in an aqueous CO_2 electrolyte. The susceptibility of experimental pipeline steels to pitting corrosion was investigated with potentiodynamic polarization and immersion corrosion tests combined with SEM/EDS analysis.

EXPERIMENTAL PROCEDURE

Materials and Chemicals

Five experimental pipeline steels were tested (H20B, H21, H22, H23, and X100). H20B, H21, H22 and H23 steel specimens were machined from an experimental plate steel manufactured at the National Energy Technology Laboratory (NETL) with a longitudinal yield strength ranging from 458 to 504 MPa. For the X100 carbon steel, the specimens were machined from an experimental pipe fabricated in the late 1990s with a yield strength of 731 MPa (in the longitudinal direction) [13]. The chemical composition of the steel pipelines is shown in Table 1. The microstructures of tested steel pipelines were obtained by etching the X100 specimens with Murakami's reagent and H20B, H21, H22, and H23 with Nital solution following the ASTM E407 standard [14]. Figure 1 shows the microstructure of tested steel pipelines. For X100 carbon steel, the microstructure consists of fine ferrite grains and bainite structure (elongated cementite [Fe_3C] particle in α-ferrite matrix) [13].

Table 1. Composition (wt.%) of Experimental Pipeline Steels

Elements	C	Mn	Al	Cr	Cu	Mo	Nb	Ni	Si	Ti+P+S	B+N	V	Ce	Fe
H20B	0.066	1.78	<0.001	0.04	0.25	0.29	0.06	0.32	0.19	0.014	0.006	0.09	0	Bal.
H21	0.056	1.77	<0.001	0.04	0.25	0.29	0.05	0.33	0.20	0.014	0.006	0.09	0.0057	Bal.
H22	0.065	1.79	0.001	0.04	0.26	0.29	0.05	0.33	0.19	0.014	0.006	0.09	0.0164	Bal.
H23	0.063	1.82	0.001	0.04	0.26	0.29	0.06	0.33	0.19	0.014	0.006	0.09	0.0263	Bal.
X100	0.085	1.69	0.029	0.19	0.14	0.17	0.047	0.24	0.26	0.031	<0.0015	-	-	Bal.

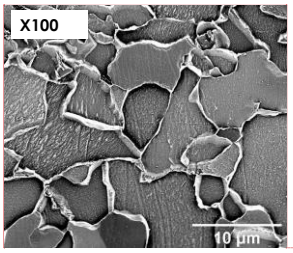


Figure 1. Microstructure of pipeline steels (H20B, H21, H22, H23, and X100).

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Electrochemical measurement

In this study, conventional electrochemical measurements were performed on five experimental pipeline steels (H20B, H21, H22, H23, and X100), in 3.5 wt.% NaCl saturated with CO_2 at 20 °C. The corrosion behavior of the pipeline steels was studied using a flat electrochemical cell with three electrodes configuration (Figure 2). Steel specimens with exposed surface area of 1 cm^2 were used as the working electrode. A saturated calomel reference electrode (SCE) was connected externally to the cell via a Luggin capillary with a porous glass tip. The counter electrode was made of a platinum mesh. The solution was deoxygenated by purging with CO_2 gas for 2 to 3 hours. After the dissolved oxygen was eliminated, the pH of the test solution reached an equilibrium pH of 3.8 to 3.9. All the corrosion tests were conducted at ambient temperature (20 °C ±0.1). Before the corrosion test, each specimen was ground using silicon

carbide paper (400 and 600 grit), rinsed with isopropanol in an ultrasonic cleaner for 2 to 3 minutes, and air dried. To keep the solution saturated with CO_2 and avoid oxygen contamination, the solution was continuously sparged with CO_2 during the corrosion tests. The measured pH of the electrolyte was approximately 3.8 ± 0.1 before and after experiments. The open-circuit potential (OCP) was monitored for 60 min for a stable signal before all the electrochemical measurements. To investigate susceptibility of experimental pipeline steels to pitting corrosion, cyclic polarization measurements were recorded from -0.5 V vs. OCP up to 1 V vs. OCP at a scan rate of 1 mV/s. Corrosion behavior of pipeline steels was monitored every hour during a total exposure time of 24 hours by measuring resistance of polarization R_p using linear polarization resistance (LPR) and electrochemical impedance spectroscopy (EIS). Polarization resistance of R_p was measured using LPR by polarizing the working electrode from -5 mV to +5 mV vs. OCP; using a scan rate of 0.125 mV/s. The Tafel value (β_a and β_c) for these experiments were determined by Tafel's extrapolation method. The corrosion current I_{corr} was calculated considering the Stern-Geary assumption (Equation 1) [15].

$$I_{corr} = \frac{1}{R_p} * \frac{\beta_a * \beta_c}{2.303(\beta_a + \beta_c)} \quad (1)$$

The corrosion rate was calculated using the following Equation [16]:

$$CR = K * \frac{I_{corr} * EW}{\rho * A} \quad (2)$$

Where *CR* is corrosion rate in mm/y; *K*: conversion factor (3,270 mm.g/A.cm. y); EW: equivalent weight (27.92 g/equivalent); *A*: area in cm^2; ρ: density of substrate (7.87 g/cm^3) I_{corr}: corrosion current in A. The EIS measurements were carried out over a frequency range of 100 kHz to 10 mHz using an AC amplitude of 10 mV (rms). Electrochemical measurements were carried out using a Gamry Reference 600+[¥] potentiostat/galvanostat. To verify the reproducibility of the results, all the electrochemical tests were repeated two to three times. The test matrix is shown in **Error! Reference source not found.**2.

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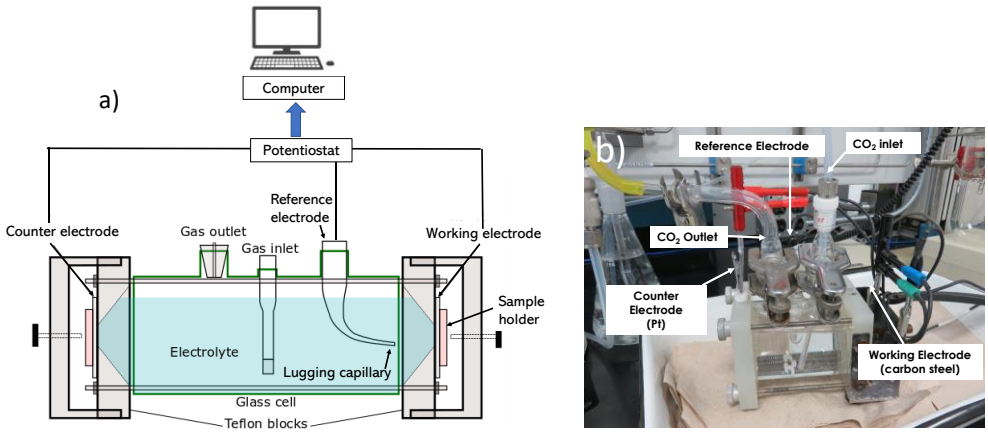


Figure 2. Experimental setup for immersion test in autoclave.

Table 2. Experimental Matrix for Electrochemical Tests

Total pressure (bar)	1
pCO_2 (bar)	0.97
Solution temperature (°C)	20 ± 0.1
Solution	0.25 L of 3.5 wt.% NaCl
pH	3.84
Tested steels	H20B, H21, H22, H23 & X100

Surface Analysis

After 24 hours of exposure to CO_2 environment, the samples were taken out, rinsed with isopropanol, dried, and stored in a desiccator for surface analyses. The exposed surface was characterized by scanning electron microscopy (SEM) and energy dispersive X-ray spectroscopy (EDS) using a using a JEOL JSM-IT700HR SEM.

RESULTS AND DISCUSSION

Potentiodynamic Polarization Curve

Potentiodynamic polarization studies on steels with and without Ce immersed in a 3.5 wt.% NaCl solution saturated with CO_2 yielded the polarization curves shown in Figure 3. Figure 3a shows anodic and cathodic polarizations curves obtained on steels after 1 hour of immersion in 3.5 wt.% NaCl solution saturated with CO_2. The tested steels showed similar behavior after 1 hour of exposure, which indicates that similar electrochemical reactions are taking place on the steel surface. The addition of Ce affected the cathodic response of the system. No significant shift of the open circuit potential was observed. As can be seen in Figure 4a, the addition of Ce to the steels did not affect the anodic current from the iron dissolution (equation 3), whereas, when the Ce is added at concentration of 0%, 0.0057% (57 ppm), and

0.0263% (263 ppm), the cathodic branch of the curves shifts to lower cathodic current densities. At the cathodic branch, the measured current density originated from the cathodic hydrogen evolution via reduction of H^+ (Equation 4) and water molecule. The protons H^+ were generated from the CO_2 dissolved in water (Equation 5) and dissociation of carbonic acid (Equation 6, 7) [17, 18]. Many studies [4, 10, 9] claimed that corrosion resistance of steel might be improved by adding Ce if the steel is exposed to atmospheric corrosion, which means that the cathodic reaction is oxygen reduction. In this case, Ce oxides will form and mitigate the cathodic sites. For CO_2 corrosion, the cathodic reaction is hydrogen evolution. Zhang et al. [19] showed that Ce can increase the rate of desorption of H atoms and ultimately increase the reduction rate of hydrogen and decrease hydrogen permeation through the steel. Therefore, accelerating the corrosion rate of the steel. As shown in Figure 3a, it seems that there is a threshold Ce concentration (0.0164% (164 ppm)) where Ce will accelerate hydrogen evolution and therefore corrosion rate. The corrosion kinetic parameters E_{corr}, i_{corr}, and CR for high strength steels with and without Ce are shown in Table 3. Cyclic polarization curves (Figure 3b) revealed no sign of pitting corrosion. Therefore, the corrosion type of tested steels in a 3.5 wt% NaCl solution saturated with CO_2 is due to uniform corrosion.

$$Fe \Leftrightarrow Fe^{2+} + 2\,e^- \qquad \text{(Equation 3)}$$

$$2\,H^+ + 2\,e^- \Leftrightarrow H_2 \qquad \text{(Equation 4)}$$

$$CO_2 + H_2O \Leftrightarrow H_2CO_3 \qquad \text{(Equation 5)}$$

$$H_2CO_3 \Leftrightarrow HCO_3^- + H^+ \qquad \text{(Equation 6)}$$

$$HCO_3^- \Leftrightarrow CO_3^{2-} + H^+ \qquad \text{(Equation 7)}$$

Table 3. Corrosion Kinetic Parameters E_{corr}, i_{corr}, and CR for the High-Strength Steels Immersed in a 3.5 wt% NaCl Solution Saturated with CO_2 at 20 °C.

Alloys	E_{corr} (V vs.SCE)	i_{corr} (A/m^2)	β_a (V/decade)	β_c (V/decade)	CR (mm/y)
H20B	-0.709 ±0.0	$1.8*10^{-5}$ ± 0.02	0.029 ± 0.003	0.097 ± 0.04	0.206 ± 0.02
H21	-0.716 ± 0.006	$2.3*10^{-5}$ ± 0.02	0.037± 0.005	0.25 ± 0.02	0.366 ± 0.02
H22	-0.723 ± 0.016	$2.6*10^{-5}$ ± 0.08	0.048 ± 0.003	0.13 ± 0.08	0.499 ± 0.08
H23	-0.712 ± 0.0	$3.9*10^{-5}$ ± 0.05	0.037± 0.003	0.179 ± 0.02	0.449 ± 0.05
X100	-0.724 ± 0.008	$5.35*10^{-5}$ ± 0.09	0.05± 0.017	0.165 ± 0.09	0.62 ± 0.09

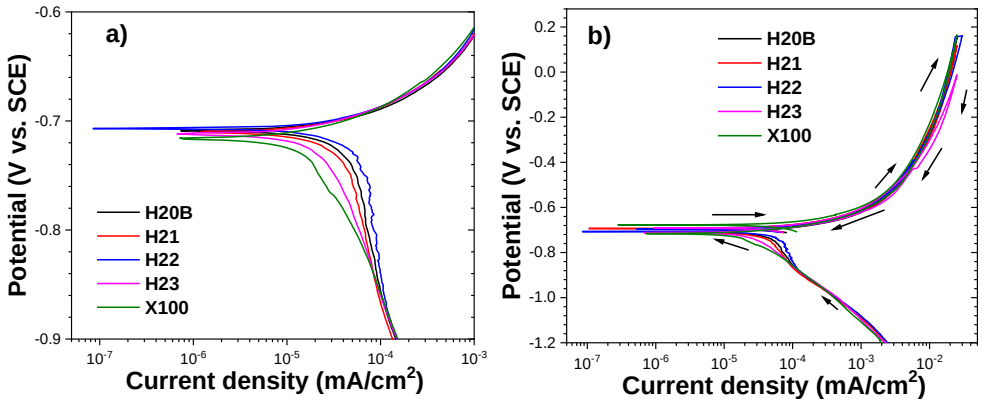


Figure 3. Potentiodynamic polarization scans obtained on steels with and without Ce immersed in a 3.5 wt.% NaCl solution saturated with CO_2 at 20 °C, and 1 bar total pressure; a). polarization curves. b). cyclic polarization curves.

Linear Polarization Resistance Corrosion Rate

The LPR and OCP measurements were carried out to study the effect of Ce on the corrosion behavior of tested steel after 24 hours of immersion in 3.5% NaCl solution. The results are shown in Figure 4. The corrosion rate for the steel without Ce (H20B) is obviously lower than that of the steel with Ce. The corrosion rates of steel increased with the increase of Ce content in steel when Ce content was in the range of 0.0%–0.0164% (0–164 ppm). Insignificant changes in corrosion rates were observed with Ce content in the range of 0.0164%–0.0263% (164–263 ppm). Steel X100 revealed a higher corrosion rate (~1 mmy) compared to the steels with and without Ce. Therefore, the corrosion resistance of the steel without Ce (H20B, 0.16 mm/y) is obviously better than that of the steel with Ce and X100 (~1 mm/y) (Figure 4a).

As shown in Figure 4, it seems that there is a threshold Ce concentration (0.0164%) whereby Ce will accelerate corrosion rate. Under the CO_2 aqueous environment, the main cathodic reaction is a reduction of H^+ and evolution of H_2 (Equation 4) and the main anodic process is the dissolution of iron to Fe^{2+} (Equation 3). With time, if the saturation level of the solution is above 1, iron and carbonate ions will react, and iron carbonate will precipitate (Equation 8), which slows down dissolution of iron and provides corrosion protection. As it was explained in the previous section, Ce can increase the rate of desorption of H atoms, ultimately increasing the reduction rate of hydrogen and decreasing hydrogen permeation through the steel [19]; therefore, accelerating the corrosion rate of the steel. In addition, it was shown in literature that manganese sulfides were replaced by Ce inclusions (Ce_xS_y-Ce_2O_2S) in the steel, which may affect its corrosion behavior [5]. The OCP (Figure 5b) followed a similar trend with and without Ce. However, the OCP shifted slightly to more cathodic potential in the presence of Ce.

$$CO_3^{2-} + Fe^{2+} \Leftrightarrow FeCO_3 \qquad \text{(Equation 8)}$$

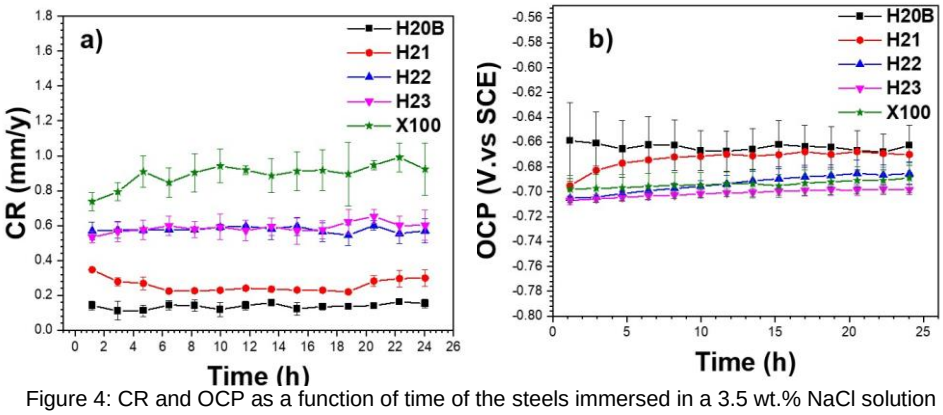


Figure 4: CR and OCP as a function of time of the steels immersed in a 3.5 wt.% NaCl solution saturated with CO_2 at 20 °C and 1 bar total pressure.

Electrochemical Impedance Spectroscopy

EIS was carried out to complement the LPR results and investigate the corrosion behavior of steel substrates under exposure to 3.5 wt.% NaCl solution saturated with CO_2. Figure 5 depicts the impedance spectra obtained on steel with and without Ce after 24 hours of exposure to 3.5 wt.% NaCl solution saturated with CO_2. All impedance spectra showed a depressed capacitive loop (Figure 5a), indicating double-layer capacitance and a one-time constant (Figure 5b) that can be assigned to the charge transfer reactions occurring on the steels. Depressed semi-circles are characteristic for heterogeneous surface roughness and the non-uniform distribution of current density on the surface [20, 21]. The low-frequency limit of the impedance decreased with increasing the Ce content, indicating decrease in polarization resistance, and increase in corrosion rate. The charge transfer resistance of steel H20B is higher (by factor 2) than the value for steel H22. Following the Ce content order, the corrosion resistance is higher for steel without Ce (HB20) and lower for the steel with 0.0263% Ce (H23). This result also showed that steel without Ce has a better corrosion resistance compared to X100 and steels with Ce. The impedance results confirm the conclusion obtained from potentiodynamic curves and LPR.

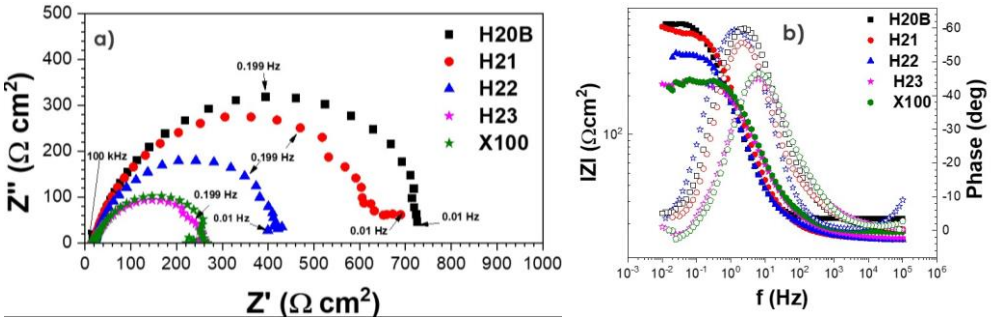


Figure 5: Nyquist diagrams (a) and Bode diagrams (b) of the steels immersed in a 3.5 wt.% NaCl solution saturated with CO_2 at 20 °C and 1 bar total pressure, after 24 hours.

Surface Characterization

Figure 6 shows SEM images of typical inclusions for the four experimental pipeline steels. As described in a concurrent study [] the inclusions are predominantly Mn-rich sulfides and Mn-Si-Al-rich oxides in the absence of Ce. The added Ce preferentially reacts with oxygen and sulfur impurities, gradually modifying the oxide and sulfide inclusions to contain higher levels of Ce (observed as bright contrast regions in Figure 6). The 0.0263% (263 ppm) Ce addition results in a nearly stoichiometric amount of Ce, given the oxygen and sulfur impurities level of the steel. As a result, virtually all the inclusions present in the 263 ppm Ce steel consisted of Ce oxysulfide, Ce_2O_2S. Figure 7 shows surface SEM and corresponding EDS maps of each of the steels after the corrosion test. The corrosion product was observed for all samples, evidenced by the oxygen maps. Carbon was also detected by EDS mapping, however in all cases quantitative EDS indicated O:C ratio much lower than 3:1, ruling out the possibility of iron carbonate ($FeCO_3$) and implying the corrosion products consisted of iron oxide or oxyhydroxide. XRD measurements of the corroded surface (data not shown) did not reveal any crystalline products; hence the precise identity of the corrosion product is unclear. For the Ce-free and 0.0057% (57 ppm) Ce samples, parts of the surface were covered by a relatively thick corrosion product (seen as O-rich regions in the EDS maps), while large regions with little or no corrosion product were also observed. Meanwhile, the 0.0164% (164 ppm) Ce sample contained some regions with relatively thick corrosion product and other regions with relatively thinner corrosion product (evidenced by less O enrichment), however corrosion-free regions such as those observed for the Ce-free and 57 ppm Ce samples were much less common. Finally, the 0.0263% (263 ppm) Ce sample and X100 were both uniformly covered by the relatively thick corrosion product, as evidenced by O enrichment uniformly across the surface. The extent of the corrosion product formation revealed in Figure 7 is in general agreement with the corrosion rates calculated from the electrochemical results, confirming the increase in Ce content was associated with a higher corrosion rate.

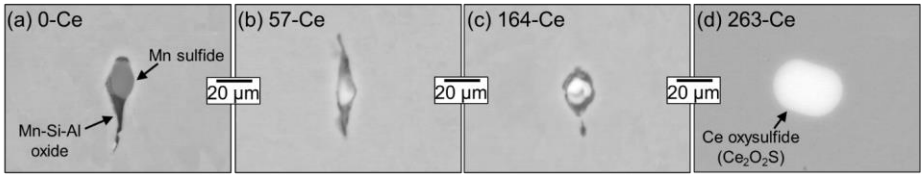


Figure 6: SEM images showing typical inclusions in the experimental pipeline steels with increasing Ce addition.

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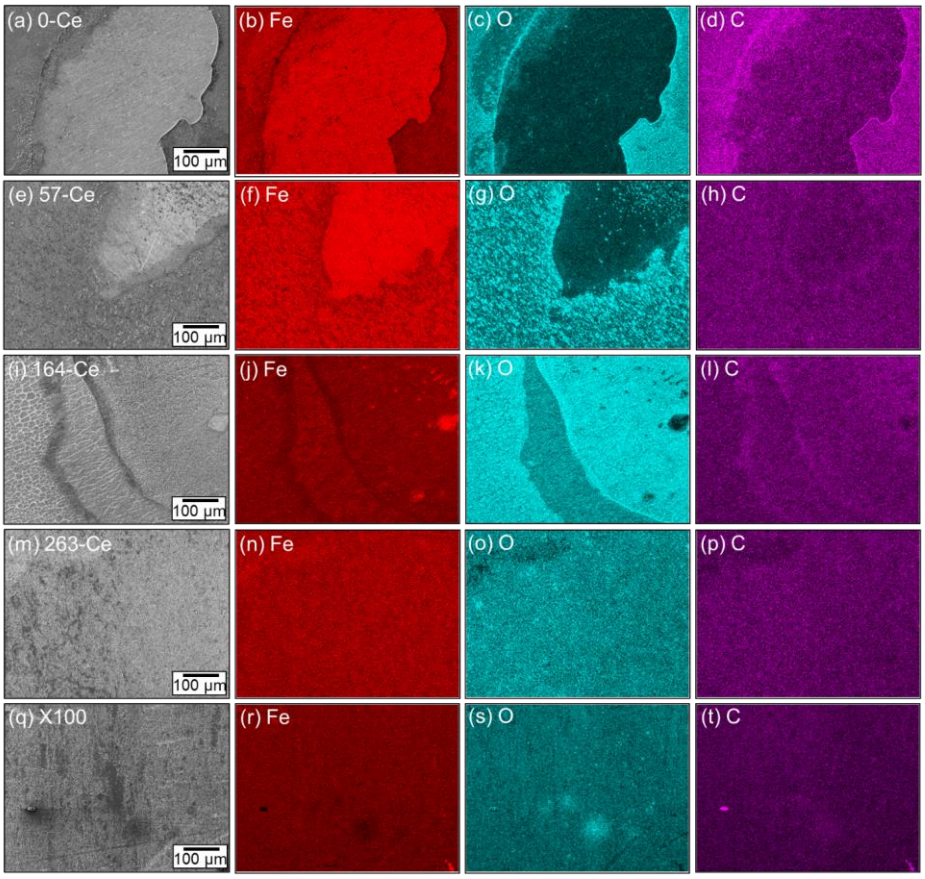


Figure 7: Secondary electron SEM images and corresponding EDS maps of typical areas of the sample surface following corrosion testing.

The corroded samples were also characterized by cross-sectional SEM. For all samples, the corrosion product was rarely observed on the surface, implying the corrosion product was not strongly bound to the steel substrate and was largely lost during preparation of the cross-sections. Nevertheless, some regions were found which retained the corrosion product on all samples except X100. For all samples, when a corrosion product layer could be found, it always exhibited a thickness of approximately 2 µm, an example of this is shown for the 263 Ce sample in Figure 8. The thickness of this corrosion product is in good agreement with the extent of corrosion expected based on the electrochemical measurements. That is, a corrosion rate of approximately 0.5 mm/y (Figure 4) corresponds to approximately 1–2 µm after the 24-hour corrosion test. The EDS maps in Figure 8 reveal little or no carbon in the corrosion product layer, again confirming the corrosion product consists of iron oxide and/or oxyhydroxide.

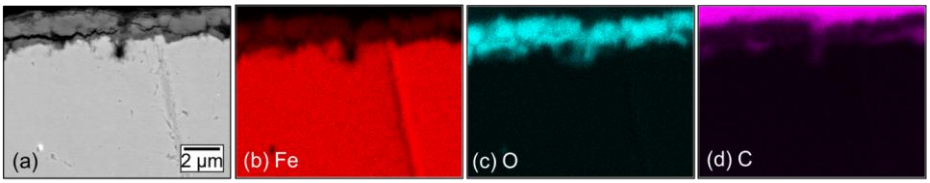


Figure 8: Cross-sectional SEM image and associated EDS maps of the corrosion product layer formed on the 0.0263% (263 ppm) Ce steel.

CONCLUSIONS

Investigations of the effect of minor Ce additions on corrosion behavior of experimental pipeline steels immersed in 3.5 wt.% NaCl solution saturated with CO_2 at 20 °C and 1 bar total pressure were performed through several electrochemical methods including potentiodynamic polarization, LPR, and EIS, followed by SEM/EDS characterization of the corrosion products. The main conclusions from this research are:

- The increase in Ce content was associated with a higher corrosion rate.
- The corrosion resistance of the steel without Ce (H20B, 0.16 mm/y) is obviously better than that of the steel with Ce and X100 (~1 mm/y).
- The added Ce preferentially reacted with oxygen and sulfur impurities, gradually modifying the oxide and sulfide inclusions to contain higher levels of Ce.
- The surface characterization revealed the formation of the iron oxide and/or oxyhydroxide layers on the corroded surfaces.
- Under the conditions tested, no localized corrosion was observed on the surface of the tested steel.

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DISCLAIMER

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